Google Sheet Cheatsheet for Data 101

Every formula in Google Sheets follows the same four rules, so learn them once and every function below becomes easier.

1. **Start with = .** The equals sign tells Sheets, "Hey, do some math for me!" Without it, Sheets just shows your words. 2. **Cells have addresses.** Column letter first, row number second. `D2` means column D, row 2. Think of it like a seat in a cinema: Row 2, Seat D.
3. **A range is a group of cells.** `D2:D7` means "from D2 all the way down to D7". The colon `:` means "to". 4. **Words go in "double quotes".** Numbers don't need quotes. `"Lemon"` is a word; `20` is a number.

Tip: Press `F4` after typing a cell address to lock it (`$D$2`). The `$` signs say "don't move this when I copy the formula".

Our practice sheet: the Lemonade Stand

All the examples in this cheatsheet use this tiny sheet. Type it in so you can follow along. Row 1 holds the headings.

	A: Seller	B: City	C: Flavor	D: Cups Sold	E: Price (₹)
2	Asha	Chennai	Lemon	10	20
3	Ben	Mumbai	Mango	5	30
4	Asha	Chennai	Mango	8	30
5	Chitra	Delhi	Lemon	12	20
6	Ben	Mumbai	Lemon	7	20
7	Dev	Delhi	Mango	(empty)	30

These six functions answer the simplest questions about a column of numbers. They all look the same: the function name, then the range in brackets.

SUM: add everything up

```
=SUM(range)
```

Like you're 10: You have a pile of coins. SUM counts all of them and tells you the total.

Example: How many cups did we sell in total?

```
=SUM(D2:D7)
```

Result: **42** (10 + 5 + 8 + 12 + 7; the empty cell is skipped).

AVERAGE: the fair share

```
=AVERAGE(range)
```

Like you're 10: Imagine sharing all the sweets equally so everyone gets the same amount. That equal amount is the average.

Example: On average, how many cups did each sale have?

```
=AVERAGE(D2:D7)
```

Result: **8.4** (42 divided by 5). Empty cells are ignored, so Dev's empty row does not pull the average down to 7. **MIN and**

MAX: the smallest and the biggest

```
=MIN(range)
=MAX(range)
```

Like you're 10: Line up your classmates by height. MIN picks the shortest one. MAX picks the tallest one. **Example:**

```
=MIN(D2:D7)
=MAX(D2:D7)
```

Results: **5** and **12**.

COUNT: how many numbers?

```
=COUNT(range)
```

Like you're 10: COUNT only counts the cells that hold **numbers**. Words, and empty cells, don't get counted. **Example:** How many rows have a number of cups sold?

```
=COUNT(D2:D7)
```

Result: **5** (Dev's cell is empty). Try `=COUNT(A2:A7)` too. It gives **0**, because names are words, not numbers. **COUNTA:**

how many cells are not empty?

```
=COUNTA(range)
```

Like you're 10: COUNTA counts **anything** that has something written in it: numbers, words, or dates. The "A" can remind you of "All".

Example: How many sellers are listed?

```
=COUNTA(A2:A7)
```

Result: **6**.

Function	Counts...	=...(A2:A7) on names	=...(D2:D7) on cups
COUNT	Only numbers	0	5
COUNTA	Anything that is not empty	6	5

Real data is messy: extra spaces, names stuck together, codes you need to cut apart. These functions are your scissors and glue.

TRIM: remove the extra spaces

```
=TRIM(text)
```

Like you're 10: Imagine a sentence wearing too many invisible spaces, like a kid in a puffy jacket. TRIM takes off the extras: no spaces at the start, no spaces at the end, and only one space between words.

Example: Cell G2 holds `" Asha Raman "` (with messy spaces).

```
=TRIM(G2)
```

Result: **Asha Raman**

Why it matters: Sheets treats `"Asha"` and `"Asha "` as different. Extra spaces silently break VLOOKUP, COUNTIFS and UNIQUE, so TRIM your data first.

LEFT, RIGHT, MID: cut out a piece of text

```
=LEFT(text, number_of_characters)
=RIGHT(text, number_of_characters)
=MID(text, start_position, number_of_characters)
```

Like you're 10: Imagine the word is a chocolate bar made of letter squares. LEFT breaks off squares from the left end. RIGHT breaks off squares from the right end. MID lets you say "start at square 3 and give me 4 squares". **Example:**

Cell H2 holds `Lemonade` .

Formula	Result	Why

`=LEFT(H2, 5)`	Lemon	First 5 letters
`=RIGHT(H2, 4)`	nade	Last 4 letters
`=MID(H2, 3, 4)`	mona	Start at letter 3, take 4 letters

CONCATENATE: glue text together

```
=CONCATENATE(text1, text2, text3, ...)
```

Like you're 10: Imagine sticking LEGO bricks together. Each piece of text is a brick, and CONCATENATE snaps them into one long piece.

Example: Join the seller and the city from row 2, with the word " from " in the middle.

```
=CONCATENATE(A2, " from ", B2)
```

Result: **Asha from Chennai**

Shortcut: The `&` sign does the same job: `=A2 & " from " & B2` .

SPLIT: break text into pieces

```
=SPLIT(text, delimiter)
```

Like you're 10: SPLIT is the opposite of gluing. You tell it where to cut (the "delimiter", such as a space or a comma) and it puts each piece in its own cell, going to the right.

Example: Cell F2 holds `Lemon,Mango,Orange` .

```
=SPLIT(F2, ",")
```

Result: **Lemon** goes in the formula cell, **Mango** in the next cell to the right, and **Orange** in the one after that. **Watch out:** Keep the cells to the right empty, or you'll see a `#REF!` error because SPLIT has no room to spread out.

```
=YEAR(date) =MONTH(date) =DAY(date)
=HOUR(time) =MINUTE(time) =SECOND(time)
```

Like you're 10: A date-and-time is like a birthday party invitation: it says the year, month, day, hour, minute and second all in one go. Each of these functions picks out just **one** piece of the invitation.

Example: Cell G2 holds the date and time `15-Oct-2026 14:35:50` .

Formula	Result	Meaning
`=YEAR(G2)`	2026	The year
`=MONTH(G2)`	10	The month number (October = 10)
`=DAY(G2)`	15	The day of the month
`=HOUR(G2)`	14	The hour, on a 24-hour clock (2 PM = 14)
`=MINUTE(G2)`	35	The minute
`=SECOND(G2)`	50	The second

Bonus helpers:

- `=TODAY()` gives today's date. `=NOW()` gives the date and the current time.
- `=DATE(2026, 10, 15)` builds a real date from three numbers.

Why it matters for MBAs: Use `=MONTH(G2)` or `=YEAR(G2)` in a helper column to group sales by month or year in a pivot table.

Watch out: If Sheets left-aligns your date, it is probably text, not a real date, and these functions will return `#VALUE!` . Real dates align to the right.

```
=UNIQUE(range)
```

Like you're 10: You have a bag of coloured marbles with many repeats. UNIQUE pulls out one marble of each colour so you can see which colours you have.

Example: Which cities are on our sheet?

```
=UNIQUE(B2:B7)
```

Result: a list that fills downward on its own: **Chennai**, **Mumbai**, **Delhi**. Each city appears once, even though some appeared twice.

Combine it: To count how many different cities there are, wrap UNIQUE inside COUNTA:

```
=COUNTA(UNIQUE(B2:B7))
```

Result: **3**.

Watch out: UNIQUE spills down into the cells below, so keep them empty. Also, `"Delhi"` and `"Delhi "` (with a trailing space) count as two different values, so clean the data with TRIM first.

IF: a yes/no question with two answers

```
=IF(test, value_if_true, value_if_false)
```

Like you're 10: "If it is raining, take an umbrella. Otherwise, wear sunglasses." IF asks a question (the test). If the answer is yes, it shows the first thing. If no, it shows the second thing.

Example: Was it a great day (10 or more cups sold)? Check row 2, where D2 = 10.

```
=IF(D2>=10, "Great day", "Keep trying")
```

Result: **Great day**. For row 3 (D3 = 5), the same formula gives **Keep trying**.

The comparison signs you can use in a test:

Sign	Means
=	is equal to
<>	is NOT equal to
>	is bigger than
<	is smaller than
>=	is bigger than or equal to
<=	is smaller than or equal to

AND: everything must be true

```
=AND(test1, test2, ...)
```

Like you're 10: "You can play outside if you finished homework AND cleaned your room." Do only one, and the answer is no. AND says TRUE only when **every** test is true.

Example: Was row 2 a Lemon sale AND 10+ cups?

```
=AND(C2="Lemon", D2>=10)
```

Result: **TRUE**. For row 3 (Mango, 5 cups) the result is **FALSE**.

OR: at least one must be true

```
=OR(test1, test2, ...)
```

Like you're 10: "You can have ice cream OR cake." Getting either one is fine. OR says TRUE when **at least one** test is true.

Example: Is the city Chennai OR Delhi?

```
=OR(B2="Chennai", B2="Delhi")
```

Result for row 2: **TRUE**. For row 3 (Mumbai): **FALSE**.

NOT: flip the answer

```
=NOT(test)
```

Like you're 10: NOT is the "opposite day" function. TRUE becomes FALSE, and FALSE becomes TRUE. **Example:**
Is row 2 NOT a Lemon sale?

```
=NOT(C2="Lemon")
```

Result: **FALSE** (because it is Lemon).

Putting them inside IF

AND, OR and NOT are usually used inside an IF so the answer is words instead of TRUE/FALSE.

```
=IF(AND(C2="Lemon", D2>=10), "Star seller", "Normal")
```

Result for row 2: **Star seller**.

IF gives only two answers. What if you want three or more, like Gold, Silver and Bronze medals? **Nested**

IF: an IF inside an IF

```
=IF(test1, answer1, IF(test2, answer2, answer3))
```

Like you're 10: Think of a bouncer with a list of questions. First question: "Did you score 10 or more?" If yes, Gold, and you're done. If no, he asks a second question: "Did you score 7 or more?" If yes, Silver. If no, Bronze.

Example: Give a medal based on cups sold in D2.

```
=IF(D2>=10, "Gold", IF(D2>=7, "Silver", "Bronze"))
```

Row	Cups	Result
2	10	Gold
6	7	Silver
3	5	Bronze

Tip: Every `IF(` you open needs its own closing `)` at the end. Count your brackets!

IFS: the tidier way

```
=IFS(test1, answer1, test2, answer2, test3, answer3, ...)
```

Like you're 10: IFS is a checklist read from top to bottom. The first question that gets a "yes" wins, and Sheets stops reading. No brackets inside brackets.

Example: Same medals, easier to read.

```
=IFS(D2>=10, "Gold", D2>=7, "Silver", TRUE, "Bronze")
```

The last pair `TRUE, "Bronze"` is a "catch-all" that means "for everything else". Without it, a value that matches none of the tests gives `#N/A` .
Order matters! Always test the **strictest** condition first (10 or more) and the loosest last. If you tested `D2>=7` first, a 12 would wrongly become Silver.

Like you're 10: Imagine a jar of mixed sweets. SUM adds up all of them. SUMIFS says, "Only add up the red ones from Delhi's jar." The "IFS" part means "only if these conditions are true". You can add as many conditions as you like, and a row must pass **all** of them.

Every function in this family is built the same way: **what to calculate**, then **pairs of (where to look, what to look for)**.

Function	Syntax
SUMIFS	`=SUMIFS(sum_range, criteria_range1, criterion1, [criteria_range2, criterion2], ...)`
COUNTIFS	`=COUNTIFS(criteria_range1, criterion1, [criteria_range2, criterion2], ...)`
COUNTUNIQUEIFS	`=COUNTUNIQUEIFS(unique_range, criteria_range1, criterion1, ...)`
AVERAGEIFS	`=AVERAGEIFS(average_range, criteria_range1, criterion1, ...)`
MINIFS	`=MINIFS(range, criteria_range1, criterion1, ...)`
MAXIFS	`=MAXIFS(range, criteria_range1, criterion1, ...)`

Memory trick: The "what to calculate" range goes **first**, except in COUNTIFS, which has nothing to add up, so it starts straight with the first condition.

Examples on the Lemonade Stand sheet

Question in plain English	Formula	Result
Total cups of Lemon sold?	`=SUMIFS(D2:D7, C2:C7, "Lemon")`	29
Total Lemon cups sold in Delhi? (two conditions)	`=SUMIFS(D2:D7, C2:C7, "Lemon", B2:B7, "Delhi")`	12
How many Mango sales rows are there?	`=COUNTIFS(C2:C7, "Mango")`	3
How many sales had 8 or more cups?	`=COUNTIFS(D2:D7, ">="&8)`	3
How many different flavors did Asha sell?	`=COUNTUNIQUEIFS(C2:C7, A2:A7, "Asha")`	2
Average cups per Lemon sale?	`=AVERAGEIFS(D2:D7, C2:C7, "Lemon")`	9.67
Smallest Lemon sale?	`=MINIFS(D2:D7, C2:C7, "Lemon")`	7
Biggest Lemon sale?	`=MAXIFS(D2:D7, C2:C7, "Lemon")`	12

How to write the "what to look for" part (the criterion)

You want...	Write it like this
Exact word	`"Lemon"` (quotes around words)
Exact number	`10`
Bigger / smaller than	`">="&8` , `"<"&5` , `"<="&8` (the sign goes inside the quotes, the number sits outside, and & joins the two)
NOT equal to	`"<>"&"Delhi"`
Words starting with L	`"L*"` (the * means "anything can follow")
Value typed in another cell (say H1)	`H1` , or `">="&H1` when using a sign

The older "single condition" cousins

SUMIF, COUNTIF and AVERAGEIF do the same job but allow only **one** condition, and the order of their ranges is slightly different.

Stick with the **-IFS** versions. They work for one condition or many, so you only need to learn one pattern.

Watch out: All the ranges must be the **same size** (for example, all `2:7`). Mixed sizes give a `#VALUE!` error.

Like you're 10: Think of a school phone directory. You know a friend's name and you want their phone number. You run your finger down the names, find your friend, then slide across to the number. Lookup functions do exactly that, in a blink.

New practice table. Type this **price list** in K1:L4. It tells us what each cup costs us to make.

	K: Flavor	L: Cost per cup (₹)
2	Lemon	8
3	Mango	14
4	Orange	12

VLOOKUP: the classic

```
=VLOOKUP(search_key, range, column_number, is_sorted)
```

Part	Meaning
search_key	What you know (the friend's name)
`range`	The table to look in. The search key must be in its **first column**
column_number	Which column of that table has your answer (1st, 2nd, 3rd...)
is_sorted	Type `FALSE` for an exact match. Almost always use `FALSE` .

Example: What is the cost per cup for the flavor in C2 (Lemon)?

```
=VLOOKUP(C2, $K$2:$L$4, 2, FALSE)
```

Result: **8**. Sheets finds "Lemon" in column K, then takes the value from the 2nd column of the table. **Why the `$` signs?** They lock the table so it does not slide down when you copy the formula to the rows below. **VLOOKUP's weak spots:**

- It can only look **to the right** of the search column.
- The column number breaks if someone inserts a new column inside your table.

XLOOKUP: the modern, friendlier one

```
=XLOOKUP(search_key, lookup_range, result_range, missing_value)
```

Part	Meaning
search_key	What you know
lookup_range	The single column or row to search in
result_range	The single column or row that holds the answer
missing_value	(optional) What to show if nothing is found, instead of an error

Like you're 10: With VLOOKUP you count columns on your fingers. With XLOOKUP you just point at two lists: "look for it here, and give me the answer from there."

Example: Same question as above, with a friendly message when the flavor is missing.

```
=XLOOKUP(C2, $K$2:$K$4, $L$2:$L$4, "Not found")
```

Result: **8**. If C2 held "Banana", you would see **Not found** instead of `#N/A` .

Looking backwards (VLOOKUP cannot do this): Which flavor costs 14?

```
=XLOOKUP(14, L2:L4, K2:K4)
```

Result: **Mango**.

	VLOOKUP	XLOOKUP
Looks to the left	No	Yes
Breaks when columns are inserted	Yes	No
Needs FALSE for exact match	Yes	No (exact is the default)
Custom message when not found	No (needs IFERROR)	Yes (4th argument)

Watch out: Lookups fail on invisible extra spaces and on numbers stored as text. TRIM your keys first.

Like you're 10: Imagine a magic highlighter pen. You tell it, "Colour every number bigger than 10 green." From then on it colours the cells by itself, even when the numbers change tomorrow.

How to open it: Select your cells, then click **Format** > **Conditional formatting**. A panel opens on the right. **Example 1:**

Highlight cells that pass a test

1. Select `D2:D7` (cups sold).
2. Under **Format cells if...**, choose **Greater than or equal to**.
3. In the value box, type `10` .
4. Under **Formatting style**, pick a green fill and click **Done**.

Result: Cups of 10 and 12 turn green. Change a number, and the colour updates on its own.

Example 2: Colour a whole row using a custom formula

This is the most powerful trick. You write a formula that gives TRUE or FALSE, and Sheets colours every row where the answer is TRUE.

1. Select `A2:E7` (the whole table, not just one column).
2. Under **Format cells if...**, choose **Custom formula is**.
3. Type:

```
=$C2="Lemon"
```

4. Pick a yellow fill and click **Done**.

Result: Every Lemon row turns yellow, from Seller to Price.

Why `$C2` ? Write the formula for the **first row** of your selection. The `$` locks column C so every cell in the row checks column C. The row number has no `$` , so each row checks its own row.

Example 3: Colour scales (a heat map)

1. Select `D2:D7` .
2. Click the **Color scale** tab at the top of the panel.
3. Choose a scale, such as white to green.

Result: The smallest number is the palest and the biggest is the darkest, so you can see high and low values at a glance. **Handy custom formulas to copy** (select the range, choose "Custom formula is", and write the formula for the first row):

Goal	Formula (for range A2:E7)

Highlight rows where cups sold are below 8	`=$D2<8`
Highlight rows for Chennai or Delhi	`=OR($B2="Chennai", $B2="Delhi")`
Highlight repeated sellers	`=COUNTIF($A$2:$A$7, $A2)>1`

Pivot Table: a summary machine

Like you're 10: You dump a big box of LEGO on the floor. You sort the bricks into piles by colour, then count each pile. A pivot table sorts your rows into piles and does the counting or adding for you, with no formulas typed.

How to build one (Lemonade Stand):

1. Click any cell in your data, or select A1:E7 (include the header row).
2. Click **Insert** > **Pivot table**, then **Create**. A new sheet opens.
3. In the right-hand panel, click **Add** next to each box:

   Rows: City

   Columns: Flavor

   Values: Cups Sold, and set **Summarize by** to **SUM**

What each box means:

Box	Like you're 10	What it does
Rows	The labels down the left side	Each unique value becomes a row
Columns	The labels across the top	Each unique value becomes a column
Values	The numbers inside the grid	What to calculate (SUM, COUNTA, AVERAGE, MIN, MAX)
Filters	A bouncer at the door	Only lets in rows that match your choice

Result:

City	Lemon	Mango	Grand Total
Chennai	10	8	18
Mumbai	7	5	12
Delhi	12	0	12
Grand Total	29	13	42

Try this: Change Summarize by from SUM to **AVERAGE** or **COUNTA** and watch the same table answer a different question. Add Seller as a second row field to break each city down further.

Good habits:

Every column in your data needs a **header** and no blank columns in between.

Pivot tables update automatically when the source data changes.

Use the month or year helper column from Level 3 to group sales by time.

Slicer: filter buttons on your sheet

Like you're 10: A slicer is a TV remote for your data. Instead of digging through menus, you press a button like "Lemon" and the pivot table instantly shows only Lemon sales.

How to add one:

1. Click inside your data or pivot table's source range.
2. Click **Data** > **Add a slicer**.
3. In the panel, choose the **column** you want to filter by, for example **Seller**, and click **Done**. 4. Click the

filter button (the small funnel) on the slicer and tick the sellers you want to see.

Result: The pivot table and charts built from the same data update instantly to show only the chosen sellers. Click the slicer's filter again and select all values to clear it.

Tip: You can add several slicers, for example one for Seller and one for City, and use them together. You can drag a slicer anywhere on the sheet and resize it.

```
=QUERY(data, "select ... where ... order by ...", headers)
```

Like you're 10: Imagine a very smart librarian. You hand over a note: "From the Lemonade table, give me just the seller and cups columns, only for Lemon, and put the biggest first." The librarian brings back exactly that. QUERY is the note you write.

Part	Meaning
`data`	The whole table, **including** the header row (A1:E7)
`"..."`	Your instructions, written inside double quotes
`headers`	How many header rows the table has. Use `1`

The three keywords (always in this order)

Keyword	Job	Like you're 10
`select`	Which **columns** to show	"Which pieces of the cake do I want?"
`where`	Which **rows** to keep	"Only the chocolate ones."
`order by`	How to **sort** the results (add `desc` for biggest first)	"Line them up from tallest to shortest."

Example 1: Pick columns with SELECT

```
=QUERY(A1:E7, "select A, D", 1)
```

Result: Only the Seller and Cups Sold columns. Use `select *` to get all columns.

Example 2: Filter rows with WHERE

```
=QUERY(A1:E7, "select A, D where C = 'Lemon'", 1)
```

Result:

Seller	Cups Sold
Asha	10
Chitra	12
Ben	7

Combine tests with `and` / `or` :

```
=QUERY(A1:E7, "select A, D where C = 'Lemon' and B = 'Delhi'", 1)
```

Result: **Chitra, 12**.

Example 3: Sort with ORDER BY

```
=QUERY(A1:E7, "select A, B, D where D >= 8 order by D desc", 1)
```

Result:

Seller	City	Cups Sold
Chitra	Delhi	12
Asha	Chennai	10
Asha	Chennai	8

This reads like a sentence: show me seller, city and cups, where cups are 8 or more, sorted from biggest to smallest.

A new practice table for dates and empty cells

On a **new sheet tab**, type this small order list into A1:C7 . Column B holds real dates, and some cells are deliberately empty.

	A: Seller	B: Order Date	C: Cups Sold
2	Asha	01-Oct-2026	10
3	Ben	05-Oct-2026	5
4	Asha	12-Oct-2026	8
5	Chitra	15-Oct-2026	12
6	Ben	(empty)	7
7	Dev	18-Oct-2026	(empty)

Example 4: Dates in WHERE

Like you're 10: QUERY does not understand a date typed as plain words. You must tell it, "Hey, what comes next is a date!" by writing the word `date` first, then the date as `'yyyy-mm-dd'` (year, month, day) in single quotes.

Orders after 10 October:

```
=QUERY(A1:C7, "select A, B where B > date '2026-10-10'", 1)
```

Result:

Seller	Order Date
Asha	12-Oct-2026
Chitra	15-Oct-2026
Dev	18-Oct-2026

Ben's empty date is not shown, because an empty cell is never "after" anything.
Orders between two dates (both days included):

```
=QUERY(A1:C7, "select A, B where B >= date '2026-10-05' and B <= date '2026-10-15'", 1)
```

Result: **Ben (05-Oct), Asha (12-Oct), Chitra (15-Oct)**.

Orders on one exact day:

```
=QUERY(A1:C7, "select A, C where B = date '2026-10-15'", 1)
```

Result: **Chitra, 12**.

Using a date typed in another cell: Say H1 holds a real date. Use `TEXT` to turn it into the `yyyy-mm-dd` style and glue it into the instruction with `&` .

```
=QUERY(A1:C7, "select A, B where B > date '" & TEXT(H1, "yyyy-mm-dd") & "'", 1)
```

Date and time together: If your column holds a date and time, write `datetime` instead of `date` : =QUERY(A1:C7,

```
"select A where B > datetime '2026-10-15 14:35:50'", 1)
```

Watch out: The column must hold **real dates**, not text. If Sheets left-aligns the dates, QUERY will not compare them properly.

Example 5: Empty cells with IS NULL and IS NOT NULL

Like you're 10: `is null` means "the box is empty, nothing inside". `is not null` means "the box has something in it". There are no quotes and no = sign: you write the words `is null` right after the column.

Find the rows where the order date is missing:

```
=QUERY(A1:C7, "select A, C where B is null", 1)
```

Result: **Ben, 7**.

Find the rows where cups sold is missing:

```
=QUERY(A1:C7, "select A, B where C is null", 1)
```

Result: **Dev, 18-Oct-2026**.

Keep only the rows where cups sold is filled in:

```
=QUERY(A1:C7, "select A, C where C is not null", 1)
```

Result:

Seller	Cups Sold
Asha	10
Ben	5
Asha	8
Chitra	12
Ben	7

Combine them, then sort: Rows where both the date and the cups are filled in, newest date first. `=QUERY(A1:C7,`

```
"select A, B, C where B is not null and C is not null order by B desc", 1)
```

Result:

Seller	Order Date	Cups Sold
Chitra	15-Oct-2026	12
Asha	12-Oct-2026	8
Ben	05-Oct-2026	5
Asha	01-Oct-2026	10

Remember: Use `is null` , never `= ''` or `= null` . Those do not work in QUERY.

Five rules that prevent most QUERY errors

1. The whole instruction lives inside **double quotes** `" "` .
2. **Words** inside the instruction go in **single quotes**: `'Lemon'` . **Numbers** need no quotes: `10` . 3. Refer to columns by their **letter** (`A` , `D`), not by their heading name.
4. Comparison signs are the same as in IF: `=` , `!=` (or `<>`), `>` , `<` , `>=` , `<=` .
5. Keep the keyword order: **select**, then **where**, then **order by**.

Using a value from another cell (say H1 holds `Lemon`):

```
=QUERY(A1:E7, "select A, D where C = '" & H1 & "'", 1)
```

The `& H1 &` part glues the cell's value into the instruction.

All syntax on one page

Function	Syntax	One-line meaning
SUM	`=SUM(range)`	Add everything
AVERAGE	`=AVERAGE(range)`	The fair share
MIN / MAX	`=MIN(range)` / `=MAX(range)`	Smallest / biggest
COUNT	`=COUNT(range)`	How many numbers
COUNTA	`=COUNTA(range)`	How many non-empty cells
TRIM	`=TRIM(text)`	Remove extra spaces
LEFT / RIGHT	`=LEFT(text, n)` / `=RIGHT(text, n)`	First / last n characters
MID	`=MID(text, start, n)`	n characters from a start position
CONCATENATE	`=CONCATENATE(a, b, ...)` or `=a & b`	Glue text together
SPLIT	`=SPLIT(text, delimiter)`	Cut text into pieces
YEAR, MONTH, DAY	`=YEAR(date)`	Pick a piece of a date
HOUR, MINUTE, SECOND	`=HOUR(time)`	Pick a piece of a time
UNIQUE	`=UNIQUE(range)`	List without repeats
IF	`=IF(test, yes, no)`	Two-way decision
AND / OR / NOT	`=AND(t1, t2)` `=OR(t1, t2)` `=NOT(t)`	All true / any true / flip
Nested IF	`=IF(t1, a, IF(t2, b, c))`	Many-way decision
IFS	`=IFS(t1, a, t2, b, TRUE, c)`	Many-way decision, tidier
SUMIFS	`=SUMIFS(sum_rng, rng1, crit1, ...)`	Add only matching rows
COUNTIFS	`=COUNTIFS(rng1, crit1, ...)`	Count matching rows
COUNTUNIQUEIFS	`=COUNTUNIQUEIFS(rng, rng1, crit1, ...)`	Count different values in matching rows
AVERAGEIFS	`=AVERAGEIFS(avg_rng, rng1, crit1, ...)`	Average of matching rows
MINIFS / MAXIFS	`=MINIFS(rng, rng1, crit1, ...)`	Smallest / biggest of matching rows
VLOOKUP	`=VLOOKUP(key, table, col, FALSE)`	Find a key, return a column to its right
XLOOKUP	`=XLOOKUP(key, look_rng, result_rng, "none")`	Find a key, return from any list
QUERY	`=QUERY(data, "select ... where ... order by ...", 1)`	Ask your data a question

When Sheets shows an error

Error	What it means	Usual fix
#NAME?	Sheets does not know the word	Check the function spelling, and put words in "quotes"
#VALUE!	Wrong kind of input, or ranges of different sizes	Check for text in a number cell and that ranges match in size
#N/A	Lookup found nothing	Check for extra spaces (use TRIM) or add a missing-value message in XLOOKUP
#REF!	The formula points to a cell that is gone, or SPLIT /	Clear the cells where results need to spill

	UNIQUE has no room	
#DIV/0!	Dividing by zero or an empty cell	Check the cell you are dividing by
#ERROR!	Sheets cannot read the formula	Look for a missing bracket or quote

The 5-second debugging checklist

1. Did the formula start with = ?
2. Do all the brackets and quote marks come in pairs?
3. Are words in "double quotes" (or 'single quotes' inside QUERY)?
4. Are all ranges the same size?
5. Are there invisible extra spaces? Clean with TRIM.